Ch 3.1: Linear Regression

Lecture 4 - CMSE 381

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Fri Sep 6, 2024

Announcements

Last time:

- 2.2 Assessing Model Accuracy

Announcements:

- Office Hours
- HW

Covered in this lecture

- Least squares coefficient estimates for linear regression
- Residual sum of squares (RSS)

Section 1

Simple Linear Regression

Setup

quantitative variable

- Predict Y on a single predictor variable X

$$Y \approx \beta_0 + \beta_1 X$$

$p = 1$

- "$\approx$" .... "is approximately modeled as"

Example

X Y

	TV	Radio	Newspaper	Sales
1	230.1	37.8	69.2	22.1
2	44.5	39.3	45.1	10.4
3	17.2	45.9	69.3	9.3
4	151.5	41.3	58.5	18.5
5	180.8	10.8	58.4	12.9
6	8.7	48.9	75	7.2
7	57.5	32.8	23.5	11.8
8	120.2	19.6	11.6	13.2
9	8.6	2.1	1	4.8
10	199.8	2.6	21.2	10.6
11	66.1	5.8	24.2	8.6

$$\text{Sales} \approx \underbrace{\beta_0}_{\text{intercept}} + \underbrace{\beta_1}_{\text{Slope}} \text{TV}$$

β_0, β_1

"coefficients" or "parameters"

Once we have a good guess

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 \text{TV}$$

fixed choice

Least squares criterion: Setup

How do we estimate the coefficients?

Given data

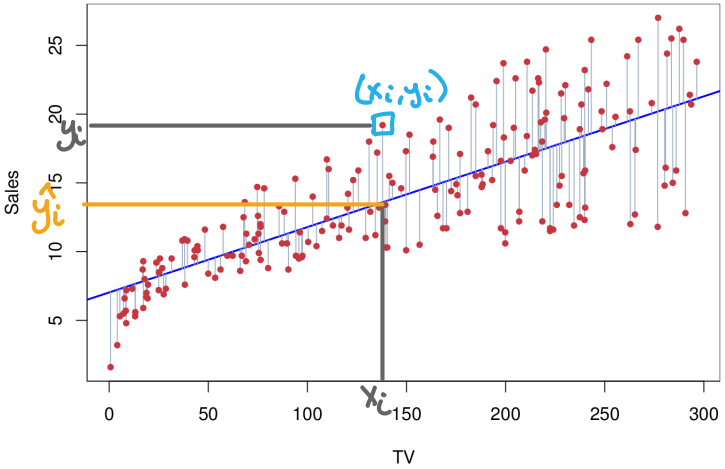
$$(x_1, y_1), \ldots, (x_n, y_n)$$

Write

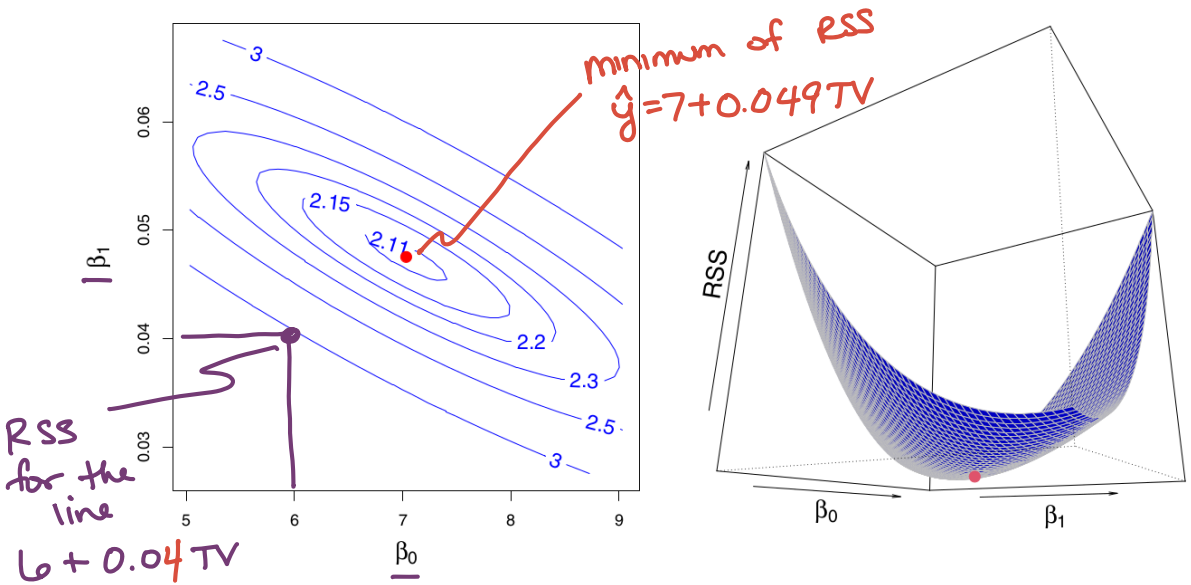
$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

The ith residual is

$$e_i = y_i - \hat{y}_i$$



Least squares criterion: RSS



$e_1 = y_1 - \hat{y}_1$

$(e_1)^2$ $(e_n)^2$

Residual sum of squares RSS is

$$RSS = e_1^2 + \cdots + e_n^2$$
$$= \sum_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$
$$= \sum (y_i - \hat{y}_i)^2$$

$$\text{sales} \approx \beta_0 + \beta_1 \text{TV}$$

$Y \approx \beta_0 + \beta_1 X$

Least squares criterion

Find β_0 and β_1 that minimize the RSS.

Least squares coefficient estimates

"Minimize RSS"

$$= \min_{\beta_0,\beta_1} \sum_i (y_i - \beta_0 - \beta_1 x_i)^2$$

fixed #'s

fixed data set

"variables"

x	y
x_i	y_i
avg $\bar{x}$	$\bar{y}$

$$\frac{\partial RSS}{\partial \beta_0} = -2\sum_i (y_i - \beta_0 - \beta_1 x_i) \boxed{= 0}$$

$$\frac{\partial RSS}{\partial \beta_1} = -2\sum_i x_i(y_i - \beta_0 - \beta_1 x_i) \boxed{= 0}$$

to find min set deriv $= 0$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

"Closed form Solution"

Can solve this!

Coding group work

☆ Only made it to here, start at the next class

Section 2

Assessing Coefficient Estimate Accuracy

Bias in estimation

Analogy with mean

- Assume a true value μ^*
- An estimate from training data $\hat{\mu}$
- The estimate is unbiased if $E(\hat{\mu} = \mu^*)$

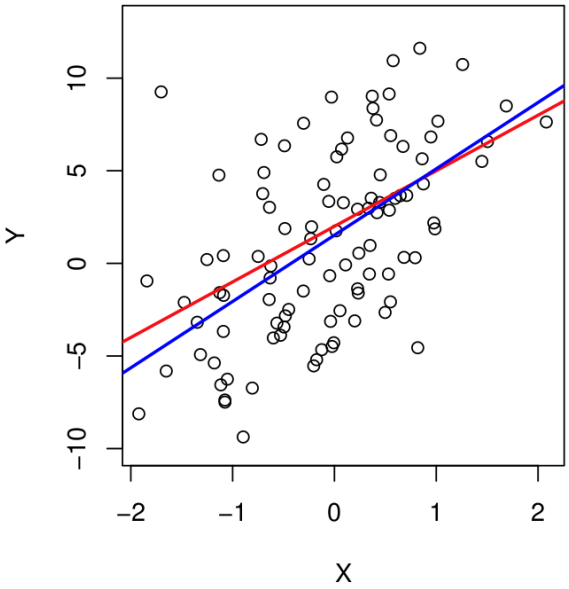
- Sample mean is unbiased for population mean:

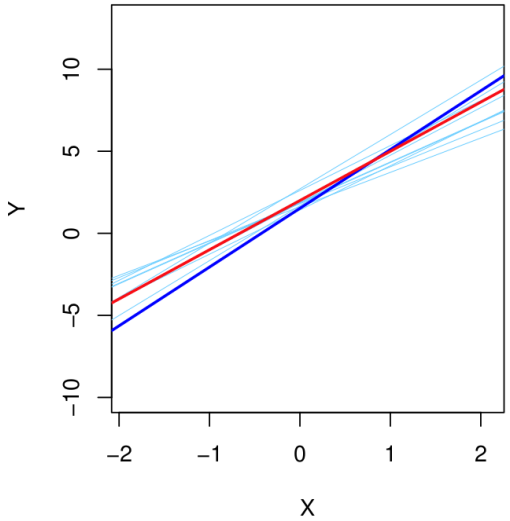
$$E(\hat{\mu}) = E\left(\frac{1}{n}\sum_i X_i\right) = \mu$$

- Standard variance estimate is biased

$$E(\hat{\sigma}^2) = E\left[\frac{1}{n}\sum_i (X_i - \overline{X})^2\right] \neq \sigma^2$$

Linear regression is unbiased





Variance in estimation

Continuing analogy with mean

- True value μ^*
- Estimate from training data $\hat{\mu}$
- Variance of sample mean
  $\mathrm{Var}(\hat{\mu}) = \mathrm{SE}(\hat{\mu})^2 = \frac{\sigma^2}{n}$

Variance of linear regression estimates

- Variance of linear regression estimates:

$$\mathrm{SE}(\hat{\beta}_0) = \sigma^2 \left[\frac{1}{n} + \frac{\overline{x}^2}{\sum_{i=1}^{n}(x_i - \overline{x})^2} \right]$$

$$\mathrm{SE}(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^{n}(x_i - \overline{x})^2}$$

where $\sigma^2 = \mathrm{Var}(\varepsilon)$

- Residual standard error is an estimate of σ

$$RSE = \sqrt{RSS/(n-2)}$$

Coding group work

Run the section titled "Simulating data"

Next time

Lec #	Date			Reading	HW
1	Mon	8/26	Intro / First day stuff / Python Review Pt 1	1	
2	Wed	8/28	What is statistical learning?	2.1	
	Fri	8/30	Class Cancelled (Dr Munch out of town)		
	Mon	9/2	No class - Labor day		
3	Wed	9/4	Assessing Model Accuracy	2.2.1, 2.2.2	
4	Fri	9/6	Linear Regression	3.1	HW #1 Due Sun 9/8
5	Mon	9/9	More Linear Regression	3.1/3.2	
6	Wed	9/11	Even more linear regression	3.2.2	
7	Fri	9/13	Probably more linear regression	3.3	Hw #2 Due

Announcements

- Homework 1
  - Due Sun, Sep 8